Three-Dimensional Nodal Homogenization for Arbitrarily Oriented and Multi-Interface Layered Media

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Abstract

We derive the nodal homogenization tensor Σ_D for a layered medium with planar symmetry direction $\hat{\boldsymbol{n}}$ at arbitrary orientation, and describe its implementation for Cartesian finite-difference grids. Both the isotropic and fully anisotropic cases are treated within the same energy-matching framework of Moskow et al. (1999), extended to three spatial dimensions. In both cases the result takes the form $\Sigma_D = \widetilde{L}^{-\top} G\, \widetilde{L}^{-1}$, where $\widetilde{L}$ encodes the discrete gradient of the local solution space and G is an energy matrix built from volume averages of the medium. For an isotropic profile the ingredients are $\widetilde{L} = [\hat{\boldsymbol{m}} \mid \hat{\boldsymbol{q}} \mid D\hat{\boldsymbol{n}}]$, with D a diagonal matrix of per-axis line averages of σ^{-1}, and $G = \operatorname{diag}(\bar{\sigma}, \bar{\sigma}, \langle \sigma^{-1} \rangle_{\text{vol}})$. For a tensor-valued profile, the columns of $\widetilde{L}$ acquire off-diagonal corrections from the off-normal conductivity components, and the upper-left block of G is replaced by the volume average of the pointwise Schur complement while G_{33} becomes the volume average of $1/\sigma_{nn}$.

Two practical computational paths are described. When the interface geometry is known analytically, $\hat{\boldsymbol{n}}$ is supplied exactly and averages are read from geometric fractions. When only a conductivity callable is available, $\hat{\boldsymbol{n}}$ is estimated by fitting a plane to interface-crossing voxel midpoints on a fine sub-grid (geometric SVD); a planarity ratio detects unreliable estimates and triggers a diagonal fallback. For tensor-valued callables the SVD uses $\frac{1}{3}\operatorname{tr}\sigma$ as a scalar proxy.

Dual cells straddling two interfaces are handled by sequential nodal homogenization: the two outer materials are homogenized first with the outer boundary normal, then the result is homogenized against the sandwiched inner material (identified by its median conductivity) with the inner boundary normal. A fallback to single-interface treatment is applied when the intermediate tensor has eigenvalues outside the physical conductivity range.

1 Setup

Let $H = [x_1^-, x_1^+] \times [x_2^-, x_2^+] \times [x_3^-, x_3^+]$ be a rectangular dual cell (node-centered box) on a Cartesian grid with grid axes $\hat{\boldsymbol{e}}_1, \hat{\boldsymbol{e}}_2, \hat{\boldsymbol{e}}_3$. We consider an isotropic medium whose conductivity varies only along a fixed direction $\hat{\boldsymbol{n}}$:

$$\sigma(\boldsymbol{x}) = \sigma(\hat{\boldsymbol{n}} \cdot \boldsymbol{x}), \tag{1}$$

where $\hat{\boldsymbol{n}}$ is a unit vector at arbitrary orientation and $\sigma : \mathbb{R} \to \mathbb{R}_{>0}$ is any measurable, bounded, and bounded-away-from-zero function. This encompasses any number of planar layers (not just one interface), as well as continuously varying profiles.

Let $\hat{\boldsymbol{m}}$ and $\hat{\boldsymbol{q}}$ be unit vectors orthogonal to $\hat{\boldsymbol{n}}$ and to each other, so that $\{\hat{\boldsymbol{m}}, \hat{\boldsymbol{q}}, \hat{\boldsymbol{n}}\}$ is an orthonormal basis.

2 Local Solution Space

Following Moskow et al. (1999), we approximate the electric potential locally by functions whose gradient satisfies the interface transmission conditions exactly at every level set of $\hat{\boldsymbol{n}} \cdot \boldsymbol{x}$: the tangential electric field and the normal current density are both constant in the $\hat{\boldsymbol{n}}$ direction. This motivates the local solution space

$$L(H) = \operatorname{span}\{1,\ \hat{\boldsymbol{m}} \cdot \boldsymbol{x},\ \hat{\boldsymbol{q}} \cdot \boldsymbol{x},\ \varphi_n(\boldsymbol{x})\},$$

where

$$\varphi_n(\boldsymbol{x}) = \int_0^{\hat{\boldsymbol{n}} \cdot \boldsymbol{x}} \frac{dt}{\sigma(t)}$$

is the "normal potential," which carries all the variability of σ across layers. Note that φ_n is well-defined for any σ satisfying (1), regardless of the number of layers or smoothness of the profile.

3 True and Discrete Gradients

True gradient. For $\phi \in L(H)$ with coefficient vector $\boldsymbol{a} = (a_1, a_2, a_3)^\top$,

$$\phi(\boldsymbol{x}) = a_1(\hat{\boldsymbol{m}} \cdot \boldsymbol{x}) + a_2(\hat{\boldsymbol{q}} \cdot \boldsymbol{x}) + a_3\, \varphi_n(\boldsymbol{x}),$$

the true gradient is

$$\nabla\phi = a_1\, \hat{\boldsymbol{m}} + a_2\, \hat{\boldsymbol{q}} + \frac{a_3}{\sigma(\hat{\boldsymbol{n}} \cdot \boldsymbol{x})}\, \hat{\boldsymbol{n}}. \tag{2}$$

The three coefficients have clear physical meaning: a_1 and a_2 are the (constant) tangential electric field components, and a_3 is the (constant) normal current density $J_n = \sigma E_n$.

Discrete gradient. The finite-difference approximation to the k-th component of $\nabla\phi$ along grid axis $\hat{\boldsymbol{e}}_k$ is the average of $\partial\phi/\partial x_k$ across the corresponding edge. Since $\phi = a_1(\hat{\boldsymbol{m}} \cdot \boldsymbol{x}) + a_2(\hat{\boldsymbol{q}} \cdot \boldsymbol{x}) + a_3\varphi_n$ and the tangential terms contribute constants $a_1\hat{m}_k$ and $a_2\hat{q}_k$, only the normal potential needs care:

$$\frac{1}{\Delta x_k} \int_{x_k^-}^{x_k^+} \frac{\partial\varphi_n}{\partial x_k}\, dx_k = \hat{n}_k \cdot \frac{1}{\Delta x_k} \int_{x_k^-}^{x_k^+} \frac{dx_k}{\sigma(\hat{\boldsymbol{n}} \cdot \boldsymbol{x})}\Big|_{\text{other coords fixed}} = \hat{n}_k\, d_k,$$

where

$$d_k = \langle \sigma^{-1} \rangle_k \ := \ \frac{1}{\Delta x_k} \int_{x_k^-}^{x_k^+} \frac{dx_k}{\sigma(\hat{\boldsymbol{n}} \cdot \boldsymbol{x})}\Big|_{\text{other coords at node}} \tag{3}$$

is the line average of σ^{-1} along axis $\hat{\boldsymbol{e}}_k$ through the node. This integral is taken at fixed coordinates perpendicular to $\hat{\boldsymbol{e}}_k$ and is well-defined for any profile $\sigma(\hat{\boldsymbol{n}} \cdot \boldsymbol{x})$. In the special case of M planar layers with conductivities $\sigma_1, \ldots, \sigma_M$ and fractional lengths $f_k^{(j)}$ along edge k, it reduces to $d_k = \sum_j f_k^{(j)}/\sigma_j$.

Assembling all three components, the discrete gradient is

$$\widetilde{\nabla}\phi = a_1\, \hat{\boldsymbol{m}} + a_2\, \hat{\boldsymbol{q}} + a_3\, D\hat{\boldsymbol{n}}, \qquad D = \operatorname{diag}(d_1, d_2, d_3), \tag{4}$$

or in matrix form

$$\widetilde{\nabla}\phi = \widetilde{L}\, \boldsymbol{a}, \qquad \widetilde{L} := [\hat{\boldsymbol{m}} \mid \hat{\boldsymbol{q}} \mid D\hat{\boldsymbol{n}}] \in \mathbb{R}^{3\times 3},$$

where the columns of $\widetilde{L}$ are $\hat{\boldsymbol{m}}$, $\hat{\boldsymbol{q}}$, and $D\hat{\boldsymbol{n}}$.

4 Energy Matching Condition

We seek an effective tensor Σ_D such that, for every $\phi, \psi \in L(H)$,

$$|H|\,\widetilde{\nabla}\phi \cdot \Sigma_D\,\widetilde{\nabla}\psi \;=\; \int_H \nabla\phi \cdot \sigma\,\nabla\psi\; dV. \tag{5}$$

Right-hand side. Using (2) and the orthonormality of $\{\hat{\boldsymbol{m}}, \hat{\boldsymbol{q}}, \hat{\boldsymbol{n}}\}$,

$$\nabla\phi \cdot \sigma\nabla\psi = \sigma\Big(a_1b_1 + a_2b_2\Big) + \frac{a_3b_3}{\sigma},$$

so

$$\int_H \nabla\phi \cdot \sigma\nabla\psi\; dV = |H|\big(\bar{\sigma}\,(a_1b_1 + a_2b_2) + \langle\sigma^{-1}\rangle_{\text{vol}}\,a_3b_3\big) = |H|\;\boldsymbol{a}^\top G\,\boldsymbol{b},$$

where

$$G = \operatorname{diag}\!\big(\bar{\sigma},\ \bar{\sigma},\ \langle\sigma^{-1}\rangle_{\text{vol}}\big), \tag{6}$$

with

$$\bar{\sigma} = \langle\sigma\rangle_{\text{vol}} := \frac{1}{|H|}\int_H \sigma(\hat{\boldsymbol{n}}\cdot\boldsymbol{x})\,dV, \qquad \langle\sigma^{-1}\rangle_{\text{vol}} = \langle\sigma^{-1}\rangle_{\text{vol}} := \frac{1}{|H|}\int_H \frac{dV}{\sigma(\hat{\boldsymbol{n}}\cdot\boldsymbol{x})},$$

the volume averages of σ and σ^{-1} over H, respectively. Both are well-defined for any measurable bounded profile (1). Note that $\langle\sigma^{-1}\rangle_{\text{vol}} = 1/\widetilde{\sigma}$, where $\widetilde{\sigma} = (\langle\sigma^{-1}\rangle_{\text{vol}})^{-1}$ is the harmonic mean of σ. The key observation is that $G_{33} = \langle\sigma^{-1}\rangle_{\text{vol}}$ is the *arithmetic* mean of the resistivity—not the harmonic mean of σ—a distinction that is critical when $D\hat{\boldsymbol{n}} \not\propto \hat{\boldsymbol{n}}$.

Left-hand side. With $\widetilde{\nabla}\phi = \widetilde{L}\boldsymbol{a}$ and $\widetilde{\nabla}\psi = \widetilde{L}\boldsymbol{b}$,

$$|H|\;\widetilde{\nabla}\phi \cdot \Sigma_D\,\widetilde{\nabla}\psi = |H|\;\boldsymbol{a}^\top\widetilde{L}^\top\Sigma_D\,\widetilde{L}\,\boldsymbol{b}.$$

Condition. Equating both sides for all $\boldsymbol{a}, \boldsymbol{b}$ gives

$$\widetilde{L}^\top\Sigma_D\,\widetilde{L} = G,$$

and solving for Σ_D:

$$\boxed{\Sigma_D = \widetilde{L}^{-\top}\,G\,\widetilde{L}^{-1}, \qquad \widetilde{L} = [\hat{\boldsymbol{m}} \mid \hat{\boldsymbol{q}} \mid D\hat{\boldsymbol{n}}], \qquad G = \operatorname{diag}\!\big(\bar{\sigma},\ \bar{\sigma},\ \langle\sigma^{-1}\rangle_{\text{vol}}\big).} \tag{7}$$

Here $\widetilde{L}^{-\top} := (\widetilde{L}^\top)^{-1} = (\widetilde{L}^{-1})^\top$.

5 Verification and Special Cases

Lemma 1 (Axis-aligned interface)**.** *For $\hat{\boldsymbol{n}} = \hat{\boldsymbol{e}}_3$, the formula* (7) *reduces to the standard arithmetic/harmonic tensor*

$$\Sigma_D = \operatorname{diag}(\bar{\sigma},\ \bar{\sigma},\ \widetilde{\sigma}).$$

Proof. Choose $\hat{\boldsymbol{m}} = \hat{\boldsymbol{e}}_1$, $\hat{\boldsymbol{q}} = \hat{\boldsymbol{e}}_2$. Then $D\hat{\boldsymbol{n}} = d_3\hat{\boldsymbol{e}}_3$ and $\widetilde{L} = \mathrm{diag}(1, 1, d_3)$. Since $\hat{\boldsymbol{n}} = \hat{\boldsymbol{e}}_3$, lines along $\hat{\boldsymbol{e}}_1$ and $\hat{\boldsymbol{e}}_2$ are parallel to all level sets of σ, so $d_1 = d_2 = \langle\sigma^{-1}\rangle_1 = \langle\sigma^{-1}\rangle_2$ each equal the pointwise value $\sigma(\boldsymbol{x}_{\text{node}})^{-1}$ (no variation along those edges). The $\hat{\boldsymbol{e}}_3$ edge varies through the full profile, giving $d_3 = \langle\sigma^{-1}\rangle_z = \langle\sigma^{-1}\rangle_{\text{vol}} = 1/\widetilde{\sigma}$ (the line average along z equals the volume average when the profile depends only on z). Therefore

$$\widetilde{L}^{-1} = \mathrm{diag}(1, 1, 1/d_3) = \mathrm{diag}(1, 1, \widetilde{\sigma}),$$

and

$$\Sigma_D = \mathrm{diag}(1, 1, \widetilde{\sigma}) \cdot \mathrm{diag}(\bar{\sigma}, \bar{\sigma}, 1/\widetilde{\sigma}) \cdot \mathrm{diag}(1, 1, \widetilde{\sigma}) = \mathrm{diag}(\bar{\sigma},\ \bar{\sigma},\ \widetilde{\sigma}). \qquad \square$$

Lemma 2 (Symmetric straddling)**.** *If $D\hat{\boldsymbol{n}}$ is parallel to $\hat{\boldsymbol{n}}$ (i.e. $D = d\,I$ for some scalar d), then $\Sigma_D = \Sigma_L := \bar{\sigma}(I - \hat{\boldsymbol{n}}\hat{\boldsymbol{n}}^\top) + \widetilde{\sigma}\,\hat{\boldsymbol{n}}\hat{\boldsymbol{n}}^\top$.*

Proof. When $D\hat{\boldsymbol{n}} = d\,\hat{\boldsymbol{n}}$ the columns of $\widetilde{L}$ are $\hat{\boldsymbol{m}}$, $\hat{\boldsymbol{q}}$, $d\hat{\boldsymbol{n}}$—an orthogonal set with squared norms $1, 1, d^2$. Hence $\widetilde{L}^{-1}$ has rows $\hat{\boldsymbol{m}}^\top$, $\hat{\boldsymbol{q}}^\top$, $\hat{\boldsymbol{n}}^\top/d$, giving $\widetilde{L}^{-\top} = [\hat{\boldsymbol{m}} \mid \hat{\boldsymbol{q}} \mid \hat{\boldsymbol{n}}/d]$. Then

$$\begin{aligned}
\Sigma_D &= [\hat{\boldsymbol{m}} \mid \hat{\boldsymbol{q}} \mid \hat{\boldsymbol{n}}/d]\ \mathrm{diag}(\bar{\sigma}, \bar{\sigma}, 1/\widetilde{\sigma}) \begin{pmatrix} \hat{\boldsymbol{m}}^\top \\ \hat{\boldsymbol{q}}^\top \\ \hat{\boldsymbol{n}}^\top/d \end{pmatrix} \\
&= \bar{\sigma}\,\hat{\boldsymbol{m}}\hat{\boldsymbol{m}}^\top + \bar{\sigma}\,\hat{\boldsymbol{q}}\hat{\boldsymbol{q}}^\top + \frac{1}{\widetilde{\sigma}d^2}\,\hat{\boldsymbol{n}}\hat{\boldsymbol{n}}^\top.
\end{aligned}$$

Since $d = \langle\sigma^{-1}\rangle_{\text{vol}}^{1/1} = 1/\widetilde{\sigma}$ (the line average equals the volume average when $D \propto I$), the last coefficient equals $\widetilde{\sigma}$, and $\hat{\boldsymbol{m}}\hat{\boldsymbol{m}}^\top + \hat{\boldsymbol{q}}\hat{\boldsymbol{q}}^\top = I - \hat{\boldsymbol{n}}\hat{\boldsymbol{n}}^\top$, giving $\Sigma_D = \Sigma_L$. $\square$

Remark 1. *Lemma 2 shows that the nodal correction is zero for any node at which $D\hat{\boldsymbol{n}} \propto \hat{\boldsymbol{n}}$—equivalently, when $d_x\hat{n}_x : d_z\hat{n}_z$ is constant along the direction of $\hat{\boldsymbol{n}}$. For a tilted interface $\hat{\boldsymbol{n}} = [1, 0, 1]/\sqrt{2}$, this occurs when $d_x = d_z$, i.e. when the edge fractions in the x- and z-directions are equal. The correction is non-trivial only for nodes where the interface cuts those edges asymmetrically.*

6 Relation to the Two-Dimensional Formula

Moskow et al. (1999) derive, for the two-dimensional isotropic case with normal $\hat{\boldsymbol{n}}$ and single tangential $\hat{\boldsymbol{m}}$, the tensor

$$\Sigma_D^{(2)} = K\,\Sigma_L^{(2)}\,K^\top, \qquad K = L^{-1},$$

where L is a 2×2 matrix (their eq. (A.21)) and $\Sigma_L^{(2)} = \mathrm{diag}(\bar{\sigma}, \widetilde{\sigma})$ in the $(\hat{\boldsymbol{m}}, \hat{\boldsymbol{n}})$ basis.

In 3D the same energy-matching argument gives an identical structure. The difference is that in 3D there are *two* tangential directions, each contributing an $\bar{\sigma}$ term to the diagonal of G. Writing Σ_L for the standard 3×3 arithmetic/harmonic tensor and noting that $G = \widetilde{L}^\top\Sigma_L\,\widetilde{L}$ holds only when $\widetilde{L}$ is orthogonal (i.e. $D\hat{\boldsymbol{n}} \propto \hat{\boldsymbol{n}}$), one sees that the 3D formula (7) is not generally equivalent to $K\Sigma_L K^\top$ with $K = \widetilde{L}^{-1}$—it is the direct energy-matching result, which happens to coincide with $K\Sigma_L K^\top$ only in the symmetric case.

7 Anisotropic Extension

We now extend the derivation to the case where the conductivity varies along $\hat{\boldsymbol{n}}$ as a *tensor-valued* profile: $\sigma = \sigma(\hat{\boldsymbol{n}} \cdot \boldsymbol{x})$, where for each $t \in \mathbb{R}$ the value $\sigma(t)$ is a symmetric positive-definite matrix. This includes any number of planar layers as well as continuously varying anisotropy. We write

$$\sigma_{mn}(t) := \hat{\boldsymbol{m}}^\top \sigma(t) \hat{\boldsymbol{n}}, \quad \sigma_{qn}(t) := \hat{\boldsymbol{q}}^\top \sigma(t) \hat{\boldsymbol{n}}, \quad \sigma_{nn}(t) := \hat{\boldsymbol{n}}^\top \sigma(t) \hat{\boldsymbol{n}} > 0,$$

and use $t = \hat{\boldsymbol{n}} \cdot \boldsymbol{x}$ as the layering coordinate.

This section follows the same structure as Appendix A of Moskow et al. (1999), extended from 2D to 3D by the second tangential direction $\hat{\boldsymbol{q}}$.

A.1 Interface constants

For any layered anisotropic medium of the form $\sigma(\hat{\boldsymbol{n}} \cdot \boldsymbol{x})$, the interface transmission conditions at every level set $\{\hat{\boldsymbol{n}} \cdot \boldsymbol{x} = t\}$ are continuity of the tangential electric field and of the normal current density. These motivate three constants that are uniform across all layers:

$$c_1 := \hat{\boldsymbol{m}} \cdot \nabla u \quad \text{(tangential field, } \hat{\boldsymbol{m}} \text{ direction)}, \tag{8}$$

$$c_2 := \hat{\boldsymbol{q}} \cdot \nabla u \quad \text{(tangential field, } \hat{\boldsymbol{q}} \text{ direction)}, \tag{9}$$

$$c_3 := \hat{\boldsymbol{n}} \cdot (\sigma \nabla u) \quad \text{(normal current density } J_n). \tag{10}$$

A.2 Electric field at each level

At each level t, the condition $c_3 = \sigma_{mn}(t) c_1 + \sigma_{qn}(t) c_2 + \sigma_{nn}(t) E_n$ determines the normal field:

$$E_n(t) \;=\; \frac{c_3 - \sigma_{mn}(t)\, c_1 - \sigma_{qn}(t)\, c_2}{\sigma_{nn}(t)}. \tag{11}$$

The full gradient at level t is therefore

$$\nabla u\big|_t \;=\; c_1\, \hat{\boldsymbol{m}} \;+\; c_2\, \hat{\boldsymbol{q}} \;+\; E_n(t)\, \hat{\boldsymbol{n}}, \tag{12}$$

and the triple (c_1, c_2, c_3) completely characterizes the local potential.

A.3 Discrete gradient

The finite-difference approximation to $(\nabla u)_k$ along grid axis k averages (12) over the corresponding edge at fixed transverse coordinates. Because c_1, c_2 are constant, only $E_n(t)$ varies along the edge. Substituting (11) and collecting terms in c_1, c_2, c_3:

$$(\widetilde{\nabla} u)_k \;=\; c_1(\hat{m}_k - \hat{n}_k\, [D_2]_{kk}) + c_2(\hat{q}_k - \hat{n}_k\, [D_3]_{kk}) + c_3\, \hat{n}_k\, [D_1]_{kk}, \tag{13}$$

where the per-axis line averages are

$$[D_1]_{kk} \;:=\; \frac{1}{\Delta x_k} \int_{x_k^-}^{x_k^+} \frac{dx_k}{\sigma_{nn}(\hat{\boldsymbol{n}} \cdot \boldsymbol{x})} \bigg|_{x_j = x_j^{\text{node}},\, j \neq k}, \tag{14}$$

$$[D_2]_{kk} \;:=\; \frac{1}{\Delta x_k} \int_{x_k^-}^{x_k^+} \frac{\sigma_{mn}(\hat{\boldsymbol{n}} \cdot \boldsymbol{x})}{\sigma_{nn}(\hat{\boldsymbol{n}} \cdot \boldsymbol{x})}\, dx_k \bigg|_{x_j = x_j^{\text{node}},\, j \neq k}, \tag{15}$$

$$[D_3]_{kk} \;:=\; \frac{1}{\Delta x_k} \int_{x_k^-}^{x_k^+} \frac{\sigma_{qn}(\hat{\boldsymbol{n}} \cdot \boldsymbol{x})}{\sigma_{nn}(\hat{\boldsymbol{n}} \cdot \boldsymbol{x})}\, dx_k \bigg|_{x_j = x_j^{\text{node}},\, j \neq k}. \tag{16}$$

Each is a 1D integral along grid axis k through the node, with the two transverse coordinates held fixed at their node values. Because $\hat{\boldsymbol{n}} \cdot \boldsymbol{x} = \hat{n}_k x_k + \sum_{j \neq k} \hat{n}_j x_j^{\text{node}}$, the integrand is a function of x_k alone, so the integrals are well-defined for any measurable profile. This is the exact anisotropic analogue of d_k in (3). Thus D_1 is the per-axis line average of $1/\sigma_{nn}$; D_2 and D_3 are the line averages of the ratios σ_{mn}/σ_{nn} and σ_{qn}/σ_{nn}. Assembling all three Cartesian components:

$$\widetilde{\nabla} u \;=\; \widetilde{L}\,\boldsymbol{c}, \qquad \widetilde{L} \;:=\; \big[\,\hat{\boldsymbol{m}} - D_2\hat{\boldsymbol{n}} \;\big|\; \hat{\boldsymbol{q}} - D_3\hat{\boldsymbol{n}} \;\big|\; D_1\hat{\boldsymbol{n}}\,\big], \qquad \boldsymbol{c} = (c_1, c_2, c_3)^\top, \tag{17}$$

where $D_j\hat{\boldsymbol{n}}$ denotes the vector with k-th entry $[D_j]_{kk}\hat{n}_k$.

Remark 2 (2D reduction)**.** *In 2D with a single tangential direction $\hat{\boldsymbol{m}}$, there is no c_2 or D_3, and D_2 reduces to the 2×2 diagonal matrix of line averages of σ_{mn}/σ_{nn}. Then $\widetilde{L} = [\hat{\boldsymbol{m}} - D_2\hat{\boldsymbol{n}} \mid D_1\hat{\boldsymbol{n}}]$, matching eq. (A.8) of Moskow et al. (1999).*

A.4 Pointwise Schur complement and energy decomposition

At each level t, define the *pointwise Schur complement* of $\sigma(t)$ with respect to $\hat{\boldsymbol{n}}$:

$$S(t) \;:=\; \sigma(t) \;-\; \frac{(\sigma(t)\hat{\boldsymbol{n}})\,(\hat{\boldsymbol{n}}^\top\sigma(t))}{\sigma_{nn}(t)}. \tag{18}$$

$S(t)$ is symmetric positive-semidefinite with null vector $\hat{\boldsymbol{n}}$ for every t. Substituting (11)–(12) into the pointwise energy density and using the symmetry of $\sigma(t)$, the normal-tangential cross-terms cancel and one obtains the clean factorization

$$\nabla u \cdot \sigma(t)\nabla u \;=\; \boldsymbol{\xi}^\top\big(\hat{\boldsymbol{m}}\;\hat{\boldsymbol{q}}\big)^\top S(t)\big(\hat{\boldsymbol{m}}\;\hat{\boldsymbol{q}}\big)\boldsymbol{\xi} \;+\; \frac{c_3^2}{\sigma_{nn}(t)}, \qquad \boldsymbol{\xi} = (c_1, c_2)^\top. \tag{19}$$

The tangential and normal contributions are *decoupled* pointwise. Integrating over H:

$$\frac{1}{|H|}\int_H \nabla u \cdot \sigma\nabla u \, dV \;=\; \boldsymbol{c}^\top G\,\boldsymbol{c}, \tag{20}$$

where the 3×3 energy matrix in the (c_1, c_2, c_3) basis is

$$G \;=\; \begin{pmatrix} \hat{\boldsymbol{m}}^\top G_{TT}\hat{\boldsymbol{m}} & \hat{\boldsymbol{m}}^\top G_{TT}\hat{\boldsymbol{q}} & 0 \\ \hat{\boldsymbol{q}}^\top G_{TT}\hat{\boldsymbol{m}} & \hat{\boldsymbol{q}}^\top G_{TT}\hat{\boldsymbol{q}} & 0 \\ 0 & 0 & G_{nn} \end{pmatrix}, \tag{21}$$

with volume averages

$$G_{TT} \;:=\; \frac{1}{|H|}\int_H S(\hat{\boldsymbol{n}} \cdot \boldsymbol{x})\, dV, \qquad G_{nn} \;:=\; \frac{1}{|H|}\int_H \frac{dV}{\sigma_{nn}(\hat{\boldsymbol{n}} \cdot \boldsymbol{x})}. \tag{22}$$

That is, G_{TT} is the *arithmetic* volume average of the pointwise Schur complement, and G_{nn} is the *arithmetic* volume average of the normal resistivity $1/\sigma_{nn}$—the anisotropic analogue of $\langle\sigma^{-1}\rangle_{\text{vol}}$ in the isotropic case. The zeros in (21) follow from $S(t)\hat{\boldsymbol{n}} = 0$ for all t, which gives $G_{TT}\hat{\boldsymbol{n}} = 0$.

A.5 Effective tensor

The energy-matching condition $\widetilde{L}^\top \Sigma_D \widetilde{L} = G$ (Section 4) gives

$$\boxed{\Sigma_D \;=\; \widetilde{L}^{-\top}\, G\, \widetilde{L}^{-1},} \tag{23}$$

with $\widetilde{L}$ and G defined by (17) and (21)–(22). Equivalently, writing $K = \widetilde{L}^{-1}$,

$$\Sigma_D \;=\; K^\top G\, K, \tag{24}$$

the exact 3D analogue of eq. (A.14) in Moskow et al. (1999). All ingredients (D_1, D_2, D_3 and the averages G_{TT}, G_{nn}) are well-defined for any measurable tensor-valued profile $\sigma(\hat{\boldsymbol{n}} \cdot \boldsymbol{x})$.

Remark 3 (Piecewise-constant reduction)**.** *For a two-layer medium with $\sigma^{(1)}, \sigma^{(2)}$ and volume fractions f, $1-f$ (and line fractions f_k, $1-f_k$), the integrals reduce to*

$$[D_1]_{kk} = \frac{f_k}{\sigma_{nn}^{(1)}} + \frac{1-f_k}{\sigma_{nn}^{(2)}}, \qquad G_{TT} = f\, S^{(1)} + (1-f)\, S^{(2)},$$

$$[D_2]_{kk} = \frac{f_k\, \sigma_{mn}^{(1)}}{\sigma_{nn}^{(1)}} + \frac{(1-f_k)\, \sigma_{mn}^{(2)}}{\sigma_{nn}^{(2)}}, \qquad G_{nn} = \frac{f}{\sigma_{nn}^{(1)}} + \frac{1-f}{\sigma_{nn}^{(2)}},$$

and similarly for D_3.

Remark 4 (Isotropic reduction)**.** *When $\sigma(t) = \sigma(t)\, I$ is scalar, $\sigma_{mn} = \sigma_{qn} = 0$ so $D_2 = D_3 = 0$. Then $\widetilde{L} = [\hat{\boldsymbol{m}} \mid \hat{\boldsymbol{q}} \mid D_1 \hat{\boldsymbol{n}}]$ with $[D_1]_{kk} = d_k$ the isotropic line average, recovering* (4)*. The Schur complement is $S(t) = \sigma(t)(I - \hat{\boldsymbol{n}}\hat{\boldsymbol{n}}^\top)$, giving $G_{TT} = \bar{\sigma}(I - \hat{\boldsymbol{n}}\hat{\boldsymbol{n}}^\top)$ and $G_{nn} = \langle \sigma^{-1} \rangle_{\mathrm{vol}}$, so $G = \mathrm{diag}(\bar{\sigma}, \bar{\sigma}, \langle \sigma^{-1} \rangle_{\mathrm{vol}})$, recovering* (6)*.*

Remark 5 (TI medium at oblique interface)**.** *For a transversely isotropic (TI) medium $\sigma = \sigma_T(I - \hat{\boldsymbol{t}}\hat{\boldsymbol{t}}^\top) + \sigma_N \hat{\boldsymbol{t}}\hat{\boldsymbol{t}}^\top$ with symmetry axis $\hat{\boldsymbol{t}}$, and with interface normal $\hat{\boldsymbol{n}}$ not aligned with $\hat{\boldsymbol{t}}$, the ratios σ_{mn}/σ_{nn} and σ_{qn}/σ_{nn} are nonzero, making D_2 and D_3 nontrivial. Omitting them leads to a k-dependent crossing position in borehole logging simulations.*

8 Summary

The three-dimensional nodal homogenization tensor for a planar interface with unit normal $\hat{\boldsymbol{n}}$ is given by

$$\Sigma_D = \widetilde{L}^{-\top}\, G\, \widetilde{L}^{-1},$$

with the following ingredients:

Quantity	Definition	Meaning
D	$\mathrm{diag}(d_1, d_2, d_3), \quad d_k = \dfrac{1}{\Delta x_k} \displaystyle\int_{x_k^-}^{x_k^+} \dfrac{dx_k}{\sigma(\hat{\boldsymbol{n}} \cdot \boldsymbol{x})}$	per-axis line avg of σ^{-1}
$\widetilde{L}$	$[\hat{\boldsymbol{m}} \mid \hat{\boldsymbol{q}} \mid D\hat{\boldsymbol{n}}], \quad \hat{\boldsymbol{m}}, \hat{\boldsymbol{q}} \perp \hat{\boldsymbol{n}}$ orthonormal	discrete-gradient map
G	$\mathrm{diag}(\bar{\sigma}, \bar{\sigma}, \langle \sigma^{-1} \rangle_{\mathrm{vol}})$	energy matrix
$\bar{\sigma}$	$\dfrac{1}{\lvert H \rvert} \displaystyle\int_H \sigma(\hat{\boldsymbol{n}} \cdot \boldsymbol{x})\, dV$	vol. avg. of σ
$\langle \sigma^{-1} \rangle_{\mathrm{vol}}$	$\dfrac{1}{\lvert H \rvert} \displaystyle\int_H \dfrac{dV}{\sigma(\hat{\boldsymbol{n}} \cdot \boldsymbol{x})}$	vol. avg. of σ^{-1}

All quantities are well-defined for any measurable profile $\sigma = \sigma(\hat{\boldsymbol{n}} \cdot \boldsymbol{x})$, with any number of layers or continuous variation. The key point is that $G_{33} = \langle \sigma^{-1} \rangle_{\mathrm{vol}}$ is the *arithmetic mean of the resistivity* over the dual-cell volume, not the harmonic mean of σ. This value arises from the integral $\int_H \sigma \, |E_n|^2 \, dV = \int_H \sigma^{-1} \, |J_n|^2 \, dV$ when $(a_1, a_2, a_3) = (0, 0, 1)$.

Anisotropic case

For a tensor-valued layered profile $\sigma = \sigma(\hat{\boldsymbol{n}} \cdot \boldsymbol{x})$, the same formula $\Sigma_D = \widetilde{L}^{-\top} G \, \widetilde{L}^{-1}$ holds with the following ingredients (see Section 7):

Quantity	Definition	Meaning
D_1	$\operatorname{diag}_k\left(\dfrac{1}{\Delta x_k} \displaystyle\int_{x_k^-}^{x_k^+} \dfrac{dx_k}{\sigma_{nn}(\hat{\boldsymbol{n}} \cdot \boldsymbol{x})} \right)$	line avg of σ_{nn}^{-1}
D_2	$\operatorname{diag}_k\left(\dfrac{1}{\Delta x_k} \displaystyle\int_{x_k^-}^{x_k^+} \dfrac{\sigma_{mn}}{\sigma_{nn}} \, dx_k \right)$	line avg of σ_{mn}/σ_{nn}
D_3	$\operatorname{diag}_k\left(\dfrac{1}{\Delta x_k} \displaystyle\int_{x_k^-}^{x_k^+} \dfrac{\sigma_{qn}}{\sigma_{nn}} \, dx_k \right)$	line avg of σ_{qn}/σ_{nn}
$\widetilde{L}$	$[\hat{\boldsymbol{m}} - D_2 \hat{\boldsymbol{n}} \mid \hat{\boldsymbol{q}} - D_3 \hat{\boldsymbol{n}} \mid D_1 \hat{\boldsymbol{n}}]$	discrete-gradient map
$S(t)$	$\sigma(t) - \dfrac{(\sigma(t)\hat{\boldsymbol{n}})(\hat{\boldsymbol{n}}^\top \sigma(t))}{\sigma_{nn}(t)}$	pointwise Schur complement
G_{TT}	$\dfrac{1}{\lvert H \rvert} \displaystyle\int_H S(\hat{\boldsymbol{n}} \cdot \boldsymbol{x}) \, dV$	vol. avg. of Schur complement
G_{nn}	$\dfrac{1}{\lvert H \rvert} \displaystyle\int_H \dfrac{dV}{\sigma_{nn}(\hat{\boldsymbol{n}} \cdot \boldsymbol{x})}$	vol. avg. of σ_{nn}^{-1}
G	$\begin{pmatrix} \hat{\boldsymbol{m}}^\top G_{TT} \hat{\boldsymbol{m}} & \hat{\boldsymbol{m}}^\top G_{TT} \hat{\boldsymbol{q}} & 0 \\ \hat{\boldsymbol{q}}^\top G_{TT} \hat{\boldsymbol{m}} & \hat{\boldsymbol{q}}^\top G_{TT} \hat{\boldsymbol{q}} & 0 \\ 0 & 0 & G_{nn} \end{pmatrix}$	energy matrix

The isotropic case is recovered when $\sigma(t) = \sigma(t) I$: then $D_2 = D_3 = 0$, $\widetilde{L}$ reduces to $[\hat{\boldsymbol{m}} \mid \hat{\boldsymbol{q}} \mid D\hat{\boldsymbol{n}}]$ with $D = D_1$, and $G = \operatorname{diag}(\bar{\sigma}, \bar{\sigma}, \langle \sigma^{-1} \rangle_{\mathrm{vol}})$.

9 Practical Application: Two Computational Paths

The derivation above assumes σ is exactly layered with known normal $\hat{\boldsymbol{n}}$ inside the dual cell H. In practice two situations arise, each handled by a distinct computational path.

9.1 Case 1 Interface geometry analytically known

When the interface is described by an explicit geometric object (e.g. a planar boundary, a cylinder, a level-set surface), the unit normal $\hat{\boldsymbol{n}}$ at each dual-cell node is available analytically.

The procedure is then direct:

1. **Evaluate material parameters.** Evaluate σ (scalar or tensor) on a fine isotropic sub-grid of spacing h_{sub} within H, using the exact interface geometry to assign the correct material to each sub-voxel.

2. **Volume fractions.** From the sub-grid masks, compute the volume fraction f of each material in H and the per-axis line fractions f_k along the central grid-axis lines.

3. **Line averages.** The per-axis line averages d_k (eq. (3)) are read directly from f_k:

$$d_k \;=\; \frac{f_k}{\sigma_1} + \frac{1-f_k}{\sigma_2}.$$

   For the anisotropic case replace σ_i by the appropriate components $\sigma_{nn}^{(i)}$, $\sigma_{mn}^{(i)}/\sigma_{nn}^{(i)}$, etc.

4. **Volume averages.** Compute $\bar{\sigma} = f\sigma_1 + (1-f)\sigma_2$ and $\langle \sigma^{-1} \rangle_{\text{vol}} = f/\sigma_1 + (1-f)/\sigma_2$ from the volume fractions.

5. **Assemble Σ_D.** Apply (7) with the known $\hat{\boldsymbol{n}}$ and the computed ingredients.

No interface-direction estimation is needed; the analytic $\hat{\boldsymbol{n}}$ is used exactly. This path is exact for planar interfaces and $O(h^2)$-accurate for curved ones (where the local planarity error over a cell of width h is $O(h^2)$).

9.2 Case 2 Only σ available as a lookup function

When $\sigma(\boldsymbol{x})$ is given only as a black-box callable (e.g. a nearest-neighbour interpolation of a fine volumetric data set), no explicit interface geometry is available. The direction $\hat{\boldsymbol{n}}$ must be *estimated* from the local variation of σ within H, and the line and volume averages computed entirely by quadrature.

Step 1 Sub-grid evaluation

Evaluate σ on a fine isotropic sub-grid of spacing h_{svd} within H, producing a 3-D array $B_{ijk} = \sigma(\boldsymbol{x}_{ijk})$. Equal spacing in all three directions is important: it eliminates sampling-induced bias in the SVD step below. The same block is reused for normal estimation, volume averages, and line averages.

Step 2 Normal estimation by SVD

When σ is tensor-valued, the scalar proxy $b_{ijk} = \frac{1}{3}\operatorname{tr} B_{ijk}$ is used in place of B_{ijk} for all steps below; the actual tensor values are retained in a parallel block for use in the homogenization integrals (Steps 3–5). This ensures that the interface-detection logic is insensitive to the anisotropy *direction* and responds only to changes in conductivity *magnitude*.

The interface normal is estimated geometrically from the positions of interface-crossing voxel pairs, not from finite differences of σ. (A gradient-based approach is biased when the grid spacing is anisotropic: a 7:1 z/x ratio inflates $\partial\sigma/\partial z$ by 7× relative to $\partial\sigma/\partial x$, producing a spurious normal near $\hat{z}$.)

1. **Binarise.** Set $\sigma_{\text{mid}} = \frac{1}{2}(\min b + \max b)$; label each sub-voxel as layer 1 ($b < \sigma_{\text{mid}}$) or layer 2 ($b \geq \sigma_{\text{mid}}$).

2. **Locate crossings.** For each of the six face-adjacency directions ($\pm x, \pm y, \pm z$), flag voxel pairs whose two members belong to different layers. Record the physical midpoint of each such pair. (Using the midpoint rather than a single voxel centre places each sample directly on the interface and avoids a "flooding" bias near corners.)

3. **Fit a plane.** Stack the midpoint coordinates into an $N \times 3$ matrix P, remove its centroid, and compute the SVD $P - \bar{P} = U\,S\,V^{\top}$. The *last* right singular vector $V_{:,3}$ (minimum-variance direction of the midpoints) is the estimated plane normal $\hat{\boldsymbol{n}}$.

4. **Sign convention.** Orient $\hat{\boldsymbol{n}}$ so that the half-space $\{\boldsymbol{x}\cdot\hat{\boldsymbol{n}} > 0\}$ has higher mean conductivity.

The confidence of the estimate is measured by the *planarity ratio*

$$r = \frac{s_3}{s_1}$$

(last to first singular value of $P - \bar{P}$). For a perfectly planar interface $s_3 \approx 0$ so $r \approx 0$; for an isotropic scatter $r \approx 1$. If $r \geq \tau$ (default $\tau = 0.7$) the block has no single dominant interface direction and the cell falls back to the diagonal (axis-aligned arithmetic/harmonic) formula. Otherwise $\hat{\boldsymbol{n}}$ is accepted and the nodal formula (7) is applied.

Step 3 Line averages by 1-D quadrature

Once $\hat{\boldsymbol{n}}$ is known, the per-axis line averages

$$d_k = \frac{1}{\Delta x_k} \int_{x_k^-}^{x_k^+} \frac{dx_k}{\sigma(\hat{\boldsymbol{n}}\cdot\boldsymbol{x})}$$

(eq. (3)) are computed by 1-D numerical quadrature along each grid axis through the node centre, with n_{line} uniformly spaced points (default 50). Similarly for the anisotropic integrands σ_{mn}/σ_{nn} and σ_{qn}/σ_{nn}.

Step 4 Volume averages by 3-D quadrature

The volume averages $\bar{\sigma} = \langle\sigma\rangle_{\text{vol}}$ and $\langle\sigma^{-1}\rangle_{\text{vol}} = \langle\sigma^{-1}\rangle_{\text{vol}}$ are computed from the same isotropic SVD sub-grid (which already provides a fine, uniform sample of H) rather than a separate coarser n_{vol}^3 grid, making them more accurate for thin-sliver cells.

Step 5 Assemble Σ_D

With the estimated $\hat{\boldsymbol{n}}$ and numerically computed $D = \operatorname{diag}(d_1, d_2, d_3)$, $\bar{\sigma}$, $\langle\sigma^{-1}\rangle_{\text{vol}}$, apply (7). Cells for which the SVD confidence is low ($r \geq \tau$) or for which H contains no interface (uniform block) receive the diagonal tensor $\Sigma_D = \operatorname{diag}(\sigma_{xx}^{\text{eff}}, \sigma_{yy}^{\text{eff}}, \sigma_{zz}^{\text{eff}})$ from axis-wise harmonic averaging.

Remark 6. *The lookup-function path is fully matrix-free: it calls $\sigma(\boldsymbol{x})$ on demand at whatever resolution is required for each dual cell, without storing a global fine-grid array. For a piecewise-constant closure the total evaluation count per node is approximately $n_{\text{svd}}^3 + 3\,n_{\text{line}}$, where $n_{\text{svd}} = \lceil \Delta x/h_{\text{svd}} \rceil \times \cdots$.*

10 Multi-Interface Cells: Sequential Nodal Homogenization

In practical geometries a dual cell may straddle *two or more* planar (or approximately planar) interfaces simultaneously. A common example in borehole logging is a cell that contains three distinct materials: borehole fluid (conductivity σ_b), invasion zone (σ_i), and undisturbed formation (σ_f), separated by the borehole wall (inner boundary, normal $\hat{\boldsymbol{n}}_{\text{in}}$) and the invasion front (outer boundary, normal $\hat{\boldsymbol{n}}_{\text{out}}$). A single interface normal cannot represent both boundaries at once.

Multi-material identification (lookup-function path)

A cell is classified as multi-material during the same pass that computes the SVD normal estimate (§9.2). The isotropic SVD sub-grid block B_{ijk} (already evaluated for normal estimation) is rounded to six significant figures and its unique conductivity levels are extracted:

$$\{\sigma^{(1)} < \sigma^{(2)} < \cdots < \sigma^{(M)}\} \;=\; \text{unique}(\text{round}(B_{ijk},\, 6))\,.$$

If $M \geq 3$ the cell is flagged as multi-material and the procedure below is applied. The rounding tolerance (six decimal places) groups physically identical conductivities that may differ by floating-point noise without merging genuinely distinct materials whose conductivities differ by more than 10^{-6}.

Determining interface structure via per-pair sub-SVDs. For each of the three pairs of adjacent conductivity levels, $(0,1)$, $(1,2)$, and the *skip pair* $(0,2)$, a masked sub-SVD is performed on the voxels belonging to that pair, using an adaptive tolerance $\delta_{ij} = 0.4 \times \min(\text{gap to nearest other level})$ to define the masks. Each sub-SVD returns an estimated normal $\hat{n}_{ij}$ and a planarity ratio r_{ij}; a pair is *resolved* if $r_{ij} < \tau$ (same threshold $\tau = 0.7$ as the single-interface case).

The resolved normals determine the cell structure:

- **All interfaces parallel** ($|\hat{n}_{01} \cdot \hat{n}_{12}| > 0.90$ and $|\hat{n}_{01} \cdot \hat{n}_{02}| > 0.90$): the three materials are co-planar strata sharing a single normal direction. The nodal formula (7) is applied once to all three materials simultaneously via a direct multi-region extension (*multiregion nodal*), using the average of the resolved normals.

- **Adjacent pairs parallel, skip pair distinct** ($\hat{n}_{01} \parallel \hat{n}_{12}$, $\hat{n}_{02} \not\parallel \hat{n}_{01}$): two interfaces with two distinct normals sequential nodal homogenization applies (see below). The *primary* normal $\hat{\boldsymbol{n}}_{\text{in}}$ is the average of $\hat{n}_{01}$ and $\hat{n}_{12}$; the *secondary* normal $\hat{\boldsymbol{n}}_{\text{out}}$ is $\hat{n}_{02}$.

- **Otherwise**: the structure is ambiguous; the cell falls back to the standard single-interface nodal path using the global SVD normal.

Sequential nodal homogenization for $M = 3$ materials

Label the three distinct materials in sorted order with conductivities $\sigma_1 < \sigma_2 < \sigma_3$ (or mean tensors $\sigma^{(j)}$ in the anisotropic case) and volume fractions f_1, f_2, f_3 in H. Let $\hat{\boldsymbol{n}}_{\text{in}}$ and $\hat{\boldsymbol{n}}_{\text{out}}$ be the inner and outer boundary normals (innermost boundary first). The *inner* (sandwiched) material is always identified as the one with the *middle* conductivity value σ_2; the two *outer* materials are σ_1 and σ_3.

Both homogenization steps apply the same 2-material nodal formula (7) there is no switch to a simpler (e.g. Backus/laminate) formula at either level:

1. **Outer homogenization.** Apply the nodal formula (7) to the two *outer* materials (σ_1, f_1) and (σ_3, f_3) with normal $\hat{\boldsymbol{n}}_{\text{out}}$, treating them as if they occupied the whole cell in proportion $f_1/(f_1+f_3)$ and $f_3/(f_1+f_3)$. Per-axis line fractions are computed from the central-axis lines of the sub-grid restricted to the union of the outer-material voxels:

$$\Sigma_D^{(13)} \;=\; \widetilde{L}_{13}^{-\top}\, G_{13}\, \widetilde{L}_{13}^{-1}.$$

2. **Inner homogenization.** Treat $\Sigma_D^{(13)}$ as one effective material and the inner material σ_2 as the other. Apply the nodal formula again with normal $\hat{\boldsymbol{n}}_{\text{in}}$ and volume fractions f_2 (inner) and f_1+f_3 (outer):

$$\Sigma_D \;=\; \widetilde{L}_{\text{in}}^{-\top}\, G_{\text{in}}\, \widetilde{L}_{\text{in}}^{-1}.$$

   Per-axis line fractions for step 2 are read from the full cell's central-axis lines (not restricted to the outer-material sub-block), so they correctly capture the inner material's extent relative to the whole dual cell.

Interface normals: exact vs. estimated

In the *exact geometry* path (§9.1), both $\hat{\boldsymbol{n}}_{\text{in}}$ and $\hat{\boldsymbol{n}}_{\text{out}}$ are supplied analytically by the interface description (e.g. the cylinder's outward normal and the dip plane's normal at the node). No sub-SVD analysis is needed.

In the *lookup-function* path (§9.2), the normals are determined by the per-pair sub-SVDs described above.

Fallback

If the sequential nodal homogenization result has a maximum eigenvalue exceeding three times the largest material conductivity (a symptom of near-degeneracy when one material has very low normal conductivity and dominates G_{nn}), the algorithm falls back to the standard single-interface 2-material nodal formula using the best-separating normal among all candidates.

Remark 7 (Nesting convention)**.** *The order of the two homogenization steps matters. The convention "outer materials first" is chosen so that the inner sandwiched material is always homogenized last, against a background that already reflects the outer structure. For DDH03-type geometries (cylindrical borehole inside a dipping formation),* $\hat{\boldsymbol{n}}_{\text{in}}$ *is the cylinder axis normal and* $\hat{\boldsymbol{n}}_{\text{out}}$ *is the formation dip-plane normal.*

Reference.

S. Moskow, V. Druskin, T. Habashy, P. Lee, and S. Davydycheva, *A finite difference scheme for elliptic equations with rough coefficients using a Cartesian grid nonconforming to interfaces*, SIAM J. Numer. Anal., 36(2):442–464, 1999.